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Master of Computer Application
Web Technology Lab BMC-251
SESSION - 2025-2026(EVEN)
SEMESTER- II

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[illegible]

Experiment 1

Objective: Create a simple webpage using HTML.

Program: Code & Screenshot

```
<!DOCTYPE html>
<html>
<head>
  <title>Simple Page</title>
  <style>
    body {
      font-family: Arial;
      text-align: center;
    }
    header {
      background: black;
      color: white;
      padding: 10px;
    }
  </style>
</head>
<body>

  <header>
    <h1>My Website</h1>
  </header>

  <h2>Welcome </h2>
  <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Cupiditate voluptatibus ut,
libero quae atque, quibusdam dolor harum iste temporibus fugit suscipit aspernatur ab
preferendis ratione eos unde neque sunt. Dicta!
.</p>

  <button>Click Me</button>

</body>
</html>
```

```
index.html > html > body > header
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>AMAZON</title>
5    <style>
6      body {
7        font-family: Arial;
8        text-align: center;
9      }
10     header {
11       background: black;
12       color: white;
13       padding: 10px;
14     }
15   </style>
16 </head>
17 <body>
18
19   <header>
20     <h1>My Website</h1>
21   </header>
22
23   <h2>Welcome </h2>
24   <p>Lorem ipsum dolor sit amet consectetur adipisicing elit. Cupiditate voluptatibus ut, libero quae atque, quibusdam dolor harum iste temporibus fugit suscipit aspernatur ab perferendis ratione eos unde neque sunt.
25   Dicta!
26   <button>Click Me</button>
27
28 </body>
29 </html>
```

Screenshot:



Result: Screenshot

Experiment 2

Objective: Create a HTML page, which has properly aligned paragraphs with image along with it.

Program: Code & Screenshot

```
<!DOCTYPE html>
<html>
<head>
  <title>Paragraph with Image</title>
  <style>
    .container {
      width: 50%;
      margin: auto;
    }
    img {
      /* float: right; */
      margin-left: 15px;
      width: 250px;
    }
    p {
      text-align: justify;
    }
  </style>
</head>
<body>
  <div class="container">
    <h2>Nature</h2>

    <p>
      Nature is the most beautiful and attractive surrounding around us.
      It provides us with everything that we need to live such as air, water, and food.
      The beauty of nature inspires us and gives us peace of mind.
    </p>
    <p>
      We should always protect nature from pollution and destruction.
      Trees, rivers, mountains, and wildlife all play an important role
      in maintaining ecological balance.
```

```
</p>
</div>
</body>
</html>
```

Screenshot:

```
index.html > html > body > div.container
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Paragraph with Image</title>
5      <style>
6          .container {
7              width: 50%;
8              margin: auto;
9          }
10         img {
11             /* float: right; */
12             margin-left: 15px;
13             width: 250px;
14         }
15         p {
16             text-align: justify;
17         }
18     </style>
19 </head>
20 <body>
21     <div class="container">
22         <h2>Nature</h2>
23
24         
27
28         <p>
29             Nature is the most beautiful and attractive surrounding around us.
30             It provides us with everything that we need to live such as air, water, and food.
31             The beauty of nature inspires us and gives us peace of mind.
32         </p>
33         <p>
34             We should always protect nature from pollution and destruction.
35             Trees, rivers, mountains, and wildlife all play an important role
36             in maintaining ecological balance.
37         </p>
38     </div>
```

Result: Screenshot

Nature



Nature is the most beautiful and attractive surrounding around us. It provides us with everything that we need to live such as air, water, and food. The beauty of nature inspires us and gives us peace of mind.

We should always protect nature from pollution and destruction. Trees, rivers, mountains, and wildlife all play an important role in maintaining ecological balance.

Experiment 3

Objective: Write a program to display list of items in different styles.

Program: Code & Screenshot

```
<!DOCTYPE html>
<html>
<head>
  <title>List Styles</title>
  <style>
    ul.square {
      list-style-type: square;
    }

    ul.circle {
      list-style-type: circle;
    }

    ol.roman {
      list-style-type: upper-roman;
    }

    ol.alpha {
      list-style-type: lower-alpha;
    }
  </style>
</head>
<body>

  <h2>Unordered List (Different Styles)</h2>

  <ul class="square">
    <li>Apple</li>
    <li>Banana</li>
    <li>Mango</li>
  </ul>

  <ul class="circle">
    <li>Car</li>
    <li>Bike</li>
    <li>Bus</li>
```


<h2>Ordered List (Different Styles)</h2>

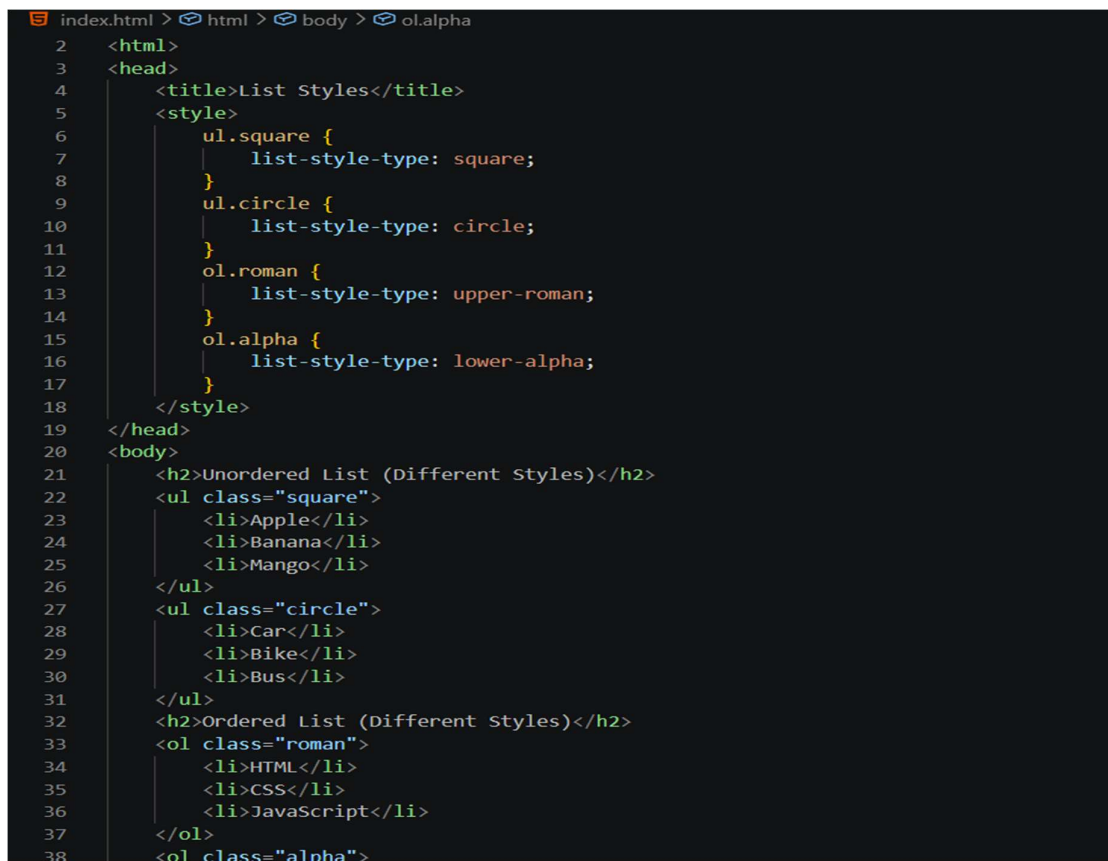
```
<ol class="roman">
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ol>
```

```
<ol class="alpha">
  <li>Red</li>
  <li>Blue</li>
  <li>Green</li>
</ol>
```

</body>

</html>

Screenshot:

A screenshot of a code editor with a dark background. The editor shows a file named 'index.html' with a breadcrumb path 'index.html > html > body > ol.alpha'. The code is as follows:

```
2 <html>
3 <head>
4   <title>List Styles</title>
5   <style>
6     ul.square {
7       list-style-type: square;
8     }
9     ul.circle {
10      list-style-type: circle;
11    }
12    ol.roman {
13      list-style-type: upper-roman;
14    }
15    ol.alpha {
16      list-style-type: lower-alpha;
17    }
18  </style>
19 </head>
20 <body>
21   <h2>Unordered List (Different Styles)</h2>
22   <ul class="square">
23     <li>Apple</li>
24     <li>Banana</li>
25     <li>Mango</li>
26   </ul>
27   <ul class="circle">
28     <li>Car</li>
29     <li>Bike</li>
30     <li>Bus</li>
31   </ul>
32   <h2>Ordered List (Different Styles)</h2>
33   <ol class="roman">
34     <li>HTML</li>
35     <li>CSS</li>
36     <li>JavaScript</li>
37   </ol>
38   <ol class="alpha">
```


Result: Screenshot

Unordered List (Different Styles)

- Apple
- Banana
- Mango
- Car
- Bike
- Bus

Ordered List (Different Styles)

- I. HTML
- II. CSS
- III. JavaScript
- a. Red
- b. Blue
- c. Green

Experiment 4

Objective: Create a Table to Show Student Details, Where Name, Roll Number, Section, Branch are included.

Program: Code & Screenshot

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Table</title>
  <style>
    table {
      border-collapse: collapse;
      width: 60%;
      margin: auto;
    }
    th, td {
      border: 1px solid black;
      padding: 8px;
      text-align: center;
    }
    th {
      background-color: lightgray;
    }
  </style>
</head>
<body>

  <h2 style="text-align:center;">Student Details</h2>

  <table>
    <tr>
      <th>Name</th>
      <th>Roll Number</th>
      <th>Section</th>
      <th>Branch</th>
    </tr>

    <tr>
      <td>Anshika</td>
      <td>101</td>
```

```
<td>A</td>
<td>CSE</td>
</tr>
```

```
<tr>
  <td>Shivansh</td>
  <td>102</td>
  <td>B</td>
  <td>IT</td>
</tr>
```

```
<tr>
  <td>Shreya</td>
  <td>103</td>
  <td>A</td>
  <td>ECE</td>
</tr>
```

```
</table>
```

```
</body>
```

```
</html>
```

Screenshot

```
index.html > html > head > style > th > td
2  <html>
3  <head>
4
5      <style>
6          table {
7              border-collapse: collapse;
8              width: 40%;
9              margin: auto;
10         }
11         th, td {
12             border: 1px solid black;
13             padding: 4px;
14             text-align: center;
15         }
16         th {
17             background-color: lightgray;
18         }
19     </style>
20 </head>
21 <body>
22     <h2 style="text-align:center;">Student Details</h2>
23     <table>
24         <tr>
25             <th>Name</th>
26             <th>Roll Number</th>
27             <th>Section</th>
28             <th>Branch</th>
29         </tr>
30         <tr>
31             <td>Anshika</td>
32             <td>101</td>
33             <td>A</td>
34             <td>CSE</td>
35         </tr>
36         <tr>
37             <td>Shivansh</td>
38             <td>102</td>
39             <td>B</td>
```

Result: Screenshot

Student Details			
Name	Roll Number	Section	Branch
Anshika	101	A	CSE
Shivansh	102	B	IT
Shreya	103	A	ECE

Experiment 5

Objective: Use frames to Include Images and Videos.

Program: Code & Screenshot

```
<!DOCTYPE html>
<html>
<head>
  <title>Frames with Image and Video</title>
</head>
<body>

  <h2 style="text-align:center;">Using Frames (iFrame)</h2>

  <!-- Frame for Image -->
  <h3>Image Frame</h3>
  <iframe src="https://encrypted-
tbn0.gstatic.com/images?q=tbn:ANd9GcQ5fUeByJSfA4Iy2TA1uJR0UYs-
LohGcfIAA&s"
    width="320" height="220">
  </iframe>

  <br><br>

  <!-- Frame for Video -->
  <h3>Video Frame</h3>
  <iframe width="320" height="220"
    src="https://www.youtube.com/embed/ysz5S6PUM-U">
  </iframe>

</body>
</html>
```

Screenshot

```
index.html > html > body > br
1  <!DOCTYPE html>
2  <html>
3  <head>
4  |   <title>Frames with Image and Video</title>
5  </head>
6  <body>
7  |
8  |   <h2 style="text-align:center;">Using Frames (iFrame)</h2>
9  |
10 |   <!-- Frame for Image -->
11 |   <h3>Image Frame</h3>
12 |   <iframe src="https://encrypted-tbn0.gstatic.com/images?q=tbn:AND9GcQ5fubEByJSfa4Iy2TA1uJR0UYs-LohGcfIAA&s"
13 |       width="320" height="220">
14 |   </iframe>
15 |
16 |   <br><br>
17 |
18 |   <!-- Frame for Video -->
19 |   <h3>Video Frame</h3>
20 |   <iframe width="320" height="220"
21 |       src="https://www.youtube.com/embed/ysz5S6PUM-U">
22 |   </iframe>
23 |
24 </body>
25 </html>
```

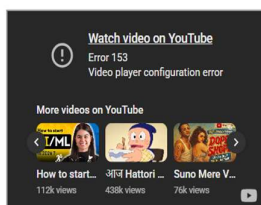
Result: Screenshot

Using Frames (iFrame)

Image Frame



Video Frame



Experiment 6

Objective: Add a Cascading Style sheet for designing the web page.

Program: Code & Screenshot

```
<!DOCTYPE html>
<html>
<head>
  <title>Colorful Cards</title>

  <style>
    body {
      font-family: Arial;
      background-color: #eef2f7;
      margin: 0;
      text-align: center;
    }

    h1 {
      margin: 20px;
      color: #2c3e50;
    }

    .container {
      display: flex;
      justify-content: center;
      gap: 20px;
      margin: 20px;
    }

    .card {
      width: 200px;
      padding: 20px;
      border-radius: 10px;
      color: white;
    }

    .card1 {
      background-color: #e74c3c;
    }

    .card2 {
      background-color: #3498db;
    }

    .card3 {
      background-color: #2ecc71;
```

```
    }
  </style>
</head>
<body>
  <h1>PICK CARDS </h1>
  <h1> Cards</h1>

  <div class="container">

    <div class="card card1">
      <h3>HTML</h3>
      <p>Structure of web pages</p>
    </div>

    <div class="card card2">
      <h3>CSS</h3>
      <p>Design and styling</p>
    </div>

    <div class="card card3">
      <h3>JavaScript</h3>
      <p>Adds interactivity</p>
    </div>

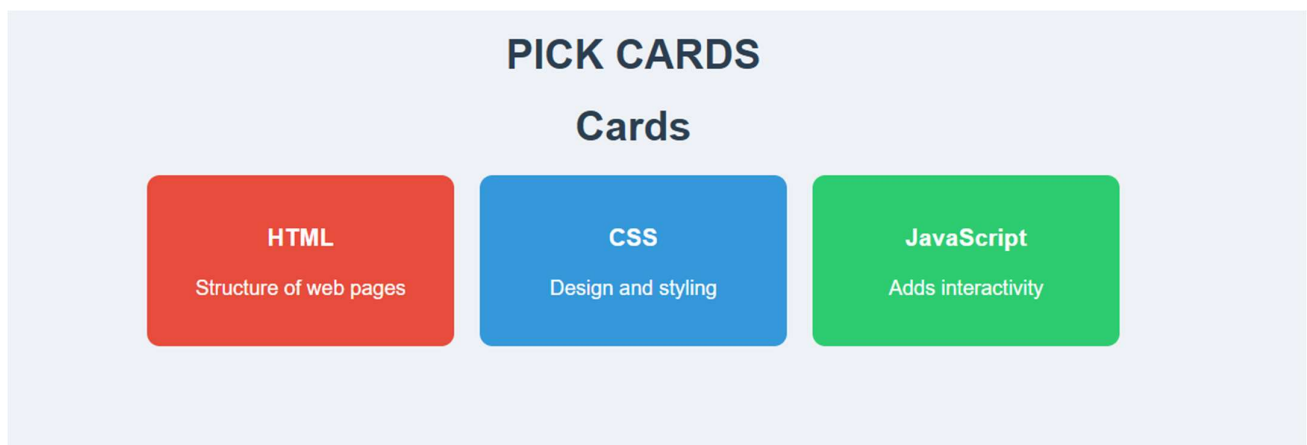
  </div>

</body>
</html>
```


Screenshot

```
index.html > html > head > style > .container
 2  <html>
 3  <head>
 6    <style>
25      .card {
27        padding: 20px;
28        border-radius: 10px;
29        color: white;
30      }
31      .card1 {
32        background-color: #e74c3c;
33      }
34      .card2 {
35        background-color: #3498db;
36      }
37      .card3 {
38        background-color: #2ecc71;
39      }
40    </style>
41  </head>
42  <body>
43    <h1>PICK CARDS </h1>
44    <h1> Cards</h1>
45
46    <div class="container">
47      <div class="card card1">
48        <h3>HTML</h3>
49        <p>Structure of web pages</p>
50      </div>
51
52      <div class="card card2">
53        <h3>CSS</h3>
54        <p>Design and styling</p>
55      </div>
56
57      <div class="card card3">
58        <h3>JavaScript</h3>
59      </div>
60    </div>
61  </body>
62 </html>
```

Result: Screenshot



Experiment – 7

Objective: Design a dynamic webpage with validation using javascript.

Program: Code & Screenshot

```
<!DOCTYPE html>
<html>
<head>
  <title>Form Validation</title>

  <style>
    body {
      font-family: Arial;
      background-color: #f4f4f4;
      text-align: center;
    }

    .form-box {
      width: 300px;
      margin: 50px auto;
      background: white;
      padding: 20px;
      border-radius: 10px;
      box-shadow: 0 0 10px lightgray;
    }

    input {
      width: 90%;
      padding: 8px;
      margin: 10px 0;
    }

    button {
      padding: 10px;
      background-color: #3498db;
      color: white;
      border: none;
      cursor: pointer;
    }

    button:hover {
      background-color: #2980b9;
    }

    .error {
      color: red;
```

```

        font-size: 14px;
    }
</style>

<script>
    function validateForm() {
        let name = document.getElementById("name").value;
        let email = document.getElementById("email").value;
        let password = document.getElementById("password").value;

        let error = "";

        if (name === "") {
            error += "Name is required<br>";
        }

        if (email === "" || !email.includes("@")) {
            error += "Valid email is required<br>";
        }

        if (password.length < 6) {
            error += "Password must be at least 6 characters<br>";
        }

        if (error !== "") {
            document.getElementById("error").innerHTML = error;
            return false;
        } else {
            alert("Form Submitted Successfully!");
            return true;
        }
    }
</script>

</head>
<body>

    <div class="form-box">
        <h2>Registration Form</h2>

        <form onsubmit="return validateForm()">
            <input type="text" id="name" placeholder="Enter Name"><br>
            <input type="text" id="email" placeholder="Enter Email"><br>
            <input type="password" id="password" placeholder="Enter Password"><br>

            <div id="error" class="error"></div>

            <button type="submit">Submit</button>
        </form>
    </div>

</body>
</html>

```

SCREENSHOT:

```
index.html > html > body
 2  <html>
 3  <head>
46  <script>
47  function validateForm() {
48
49      if (password.length < 6) {
50          error += "Password must be at least 6 characters<br>";
51      }
52
53      if (error != "") {
54          document.getElementById("error").innerHTML = error;
55          return false;
56      } else {
57          alert("Form Submitted Successfully!");
58          return true;
59      }
60  }
61  </script>
62
63 </head>
64 <body>
65
66     <div class="form-box">
67         <h2>Registration Form</h2>
68
69         <form onsubmit="return validateForm()">
70             <input type="text" id="name" placeholder="Enter Name"><br>
71             <input type="text" id="email" placeholder="Enter Email"><br>
72             <input type="password" id="password" placeholder="Enter Password"><br>
73
74             <div id="error" class="error"></div>
75
76             <button type="submit">Submit</button>
77         </form>
78     </div>
79
80 </body>
```

RESULT SCREENSHOT:

Registration Form

MarketHive

swati235@gmail.com

....

Password must be at least 6 characters

Submit

EXPERIMENT – 8

Objective: WAP using JS to demonstrate the concept of built in array methods.

Program: Code and Screenshots:

```
<!DOCTYPE html>
<html>
<head>
  <title>Array Methods</title>
</head>
<body>

<h2>JavaScript Array Methods</h2>

<p id="output"></p>

<script>
  let arr = [10, 20, 30, 40];

  // push() → adds element at end
  arr.push(50);

  // pop() → removes last element
  arr.pop();

  // shift() → removes first element
  arr.shift();

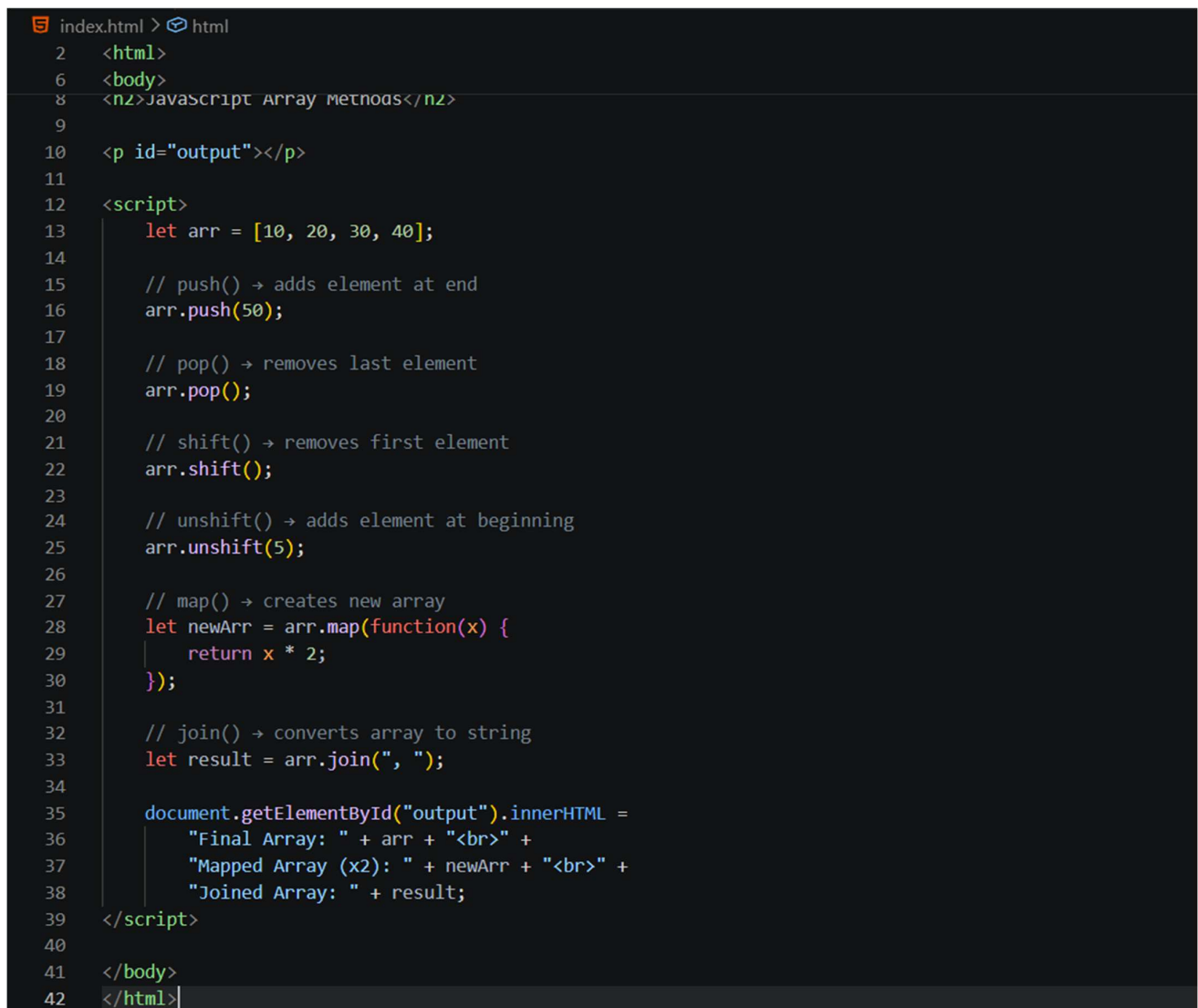
  // unshift() → adds element at beginning
  arr.unshift(5);

  // map() → creates new array
  let newArr = arr.map(function(x) {
    return x * 2;
  });

  // join() → converts array to string
  let result = arr.join(", ");
```

```
document.getElementById("output").innerHTML =  
    "Final Array: " + arr + "<br>" +  
    "Mapped Array (x2): " + newArr + "<br>" +  
    "Joined Array: " + result;  
</script>  
  
</body>  
</html>
```

SCREENSHOT:



```
index.html > html  
2  <html>  
6  <body>  
8  <nz>JavaScript Array Methods</nz>  
9  
10 <p id="output"></p>  
11  
12 <script>  
13     let arr = [10, 20, 30, 40];  
14  
15     // push() → adds element at end  
16     arr.push(50);  
17  
18     // pop() → removes last element  
19     arr.pop();  
20  
21     // shift() → removes first element  
22     arr.shift();  
23  
24     // unshift() → adds element at beginning  
25     arr.unshift(5);  
26  
27     // map() → creates new array  
28     let newArr = arr.map(function(x) {  
29         return x * 2;  
30     });  
31  
32     // join() → converts array to string  
33     let result = arr.join(", ");  
34  
35     document.getElementById("output").innerHTML =  
36         "Final Array: " + arr + "<br>" +  
37         "Mapped Array (x2): " + newArr + "<br>" +  
38         "Joined Array: " + result;  
39 </script>  
40  
41 </body>  
42 </html>
```

Result Screenshot:

JavaScript Array Methods

Final Array: 5,20,30,40

Mapped Array (x2): 10,40,60,80

Joined Array: 5, 20, 30, 40